

Product Line Profitability & Margin Performance Analysis for Nassau Candy Distributor

Abstract

In highly competitive distribution industries, sales volume alone does not accurately represent business success. Organizations must evaluate profitability, margin efficiency, and cost performance to make informed strategic decisions. This project focuses on analysing product-level profitability and margin performance for Nassau Candy Distributor using Python, PostgreSQL, and Streamlit. The objective is to identify high-performing products, detect low-margin products, evaluate division-wise profitability, and generate actionable business recommendations. Through KPI calculations, profitability analysis, Pareto analysis, and cost diagnostics, the project provides data-driven insights that support pricing optimization, cost reduction, and portfolio management decisions.

1. Introduction

Nassau Candy Distributor operates across multiple product divisions and regions. Although the organization generates substantial sales volume, management lacks visibility into which products genuinely contribute to organizational profitability.

Traditional revenue analysis often overlooks important financial indicators such as gross margin, profit contribution, and cost efficiency. This project transforms raw transactional data into meaningful business intelligence by evaluating profitability across products, divisions, and regions.

The analysis leverages Python for data processing, PostgreSQL for database analytics, and Streamlit for dashboard visualization.

2. Business Problem Statement

The organization faced several analytical challenges:

- Lack of visibility into product-level profitability
- Inability to identify high-margin products
- Difficulty detecting products with strong sales but weak profit performance
- Limited understanding of division-level financial efficiency
- Absence of structured margin risk monitoring

Without these insights, pricing, sourcing, and portfolio decisions remained reactive rather than data-driven.

3. Project Objectives

The primary objectives of this project are:

- Analyse product profitability performance
- Calculate gross margin and profitability KPIs
- Identify high-profit and low-profit products
- Compare financial performance across divisions
- Conduct Pareto analysis for revenue and profit concentration
- Evaluate cost efficiency and pricing effectiveness
- Develop an interactive business dashboard
- Generate strategic business recommendations

4. Dataset Overview

The dataset contains transactional sales information including:

Column	Description
Order ID	Unique order identifier
Order Date	Order placement date
Ship Date	Shipping date
Product Name	Product description
Division	Product category
Sales	Revenue generated
Units	Quantity sold
Gross Profit	Profit generated
Cost	Manufacturing or acquisition cost
Column	Description
Region	Sales region
State/Province	Customer location

The dataset also includes factory mapping information linking products to manufacturing locations.

5. Tools and Technologies Used

Tool	Purpose
Python	Data Cleaning & Analysis
Pandas	Data Manipulation
NumPy	Numerical Computations
PostgreSQL	Database Analytics
SQL	Business Queries
Streamlit	Interactive Dashboard
Plotly	Data Visualization
Power BI Concepts	Business Intelligence Design

6. Data Cleaning and Preprocessing

The dataset underwent extensive preprocessing before analysis. **Steps**

Performed

1. Date conversion using datetime functions
2. Handling missing values
3. Removing duplicate records
4. Validating sales and cost values
5. Standardizing product and division labels
6. Creating derived KPI columns

Example Python operations included:

- Datetime conversion

```
#Convert dates
df['ship_date'] = pd.to_datetime(df['ship_date'], format='%d-%m-%Y')
df['order_date'] = pd.to_datetime(df['order_date'])
[8]
```

- Null value handling

```
# fill missing values
df['units'] = df['units'].fillna(df['units'].median())
```

- Invalid Record removal

```
#Remove Invalid Records

df = df[df['sales'] > 0]
df = df[df['units'] > 0]

[10]
```

- Standardising Column

```
#column Standardization
df.columns=df.columns.str.lower()
df.columns=df.columns.str.replace(' ','_')
df.columns

[6]

Index(['row_id', 'order_id', 'order_date', 'ship_date', 'ship_mode',
       'customer_id', 'country/region', 'city', 'state/province',
       'postal_code', 'division', 'region', 'product_id', 'product_name',
       'sales', 'units', 'gross_profit', 'cost'],
      dtype='str')
```

This preprocessing ensured data quality and analytical reliability.

7. KPI Development

Several key performance indicators were created to evaluate profitability.

Gross Margin Percentage

Measures profitability efficiency.

```
# Gross Margin
df['gross_margin %'] = (df['gross_profit'] / df['sales']) * 100
```

Profit Per Unit

Measures profit generated per product unit sold.

```
#profit per unit
df['profit_per_unit'] = df['gross_profit'] / df['units']
✓ [75] 16ms
```

Revenue Contribution

Measures percentage contribution to total sales.

```
# Revenue Contribution
total_sales = df['sales'].sum()

df['Revenue Contribution %'] = (
    df['sales'] / total_sales
) * 100
✓ [77] 16ms
```

Profit Contribution

Measures percentage contribution to total organizational profit.

```
# Profit Contribution
total_profit = df['gross_profit'].sum()
```

Margin Volatility

Measures variation in margin performance over time.

```
#Volatility
margin_volatility = monthly_margin.groupby(
    'product_name'
)['gross_margin %'].std().reset_index()
✓ [87] 15ms
```

Cost Efficiency Ratio

Measures cost proportion relative to sales.

```
#Create Cost Efficiency KPI
df['cost_efficiency_ratio %'] = (
    df['cost'] / df['sales']
) * 100
[124]
```

8. Product Profitability Analysis

Product profitability analysis was performed to rank products according to:

- Total Gross Profit
- Gross Margin Percentage
- Revenue Contribution

Key Findings

- Several products generated high revenue but weak profitability.
- High-margin products demonstrated superior financial efficiency.
- Certain products exhibited low sales and low profit, indicating portfolio review opportunities.

The analysis classified products into:

- High Profit – High Margin

High - Profit / High - Margin Products



```
avg_profit = product_analysis['gross_profit'].mean()
avg_margin = product_analysis['gross_margin %'].mean()

high_profit_high_margin = product_analysis[
    (product_analysis['gross_profit'] > avg_profit) &
    (product_analysis['gross_margin %'] > avg_margin)
]
high_profit_high_margin
✓ [95] 44ms
```

	product_name	sales	gross_profit	gross_margin %	units
9	Wonka Bar - Fudge Mallows	24890.40	16593.60	66.67	6914
10	Wonka Bar - Milk Chocolate	26867.75	17443.37	64.92	8267
11	Wonka Bar - Nutty Crunch Surprise	23574.95	16819.95	71.35	6755
12	Wonka Bar - Triple Dazzle Caramel	28485.00	18610.20	65.33	7596
13	Wonka Bar -Scrumdiddlyumptious	27874.80	19357.50	69.44	7743

- High Sales – Low Margin

High - Sales / Low - Margin Products

```
avg_sales = product_analysis['sales'].mean()

high_sales_low_margin = product_analysis[
    (product_analysis['sales'] > avg_sales) &
    (product_analysis['gross_margin %'] < avg_margin)
]
high_sales_low_margin
```

✓ [96] 38ms

	product_name	sales	gross_profit	gross_margin %	units
--	--------------	-------	--------------	----------------	-------

- Low Sales – Low Profit

Low - Sales / Low - Profit Products

```
low_sales_low_profit = product_analysis[
    (product_analysis['sales'] < avg_sales) &
    (product_analysis['gross_profit'] < avg_profit)
]
low_sales_low_profit
```

✓ [97] 24ms

	product_name	sales	gross_profit	gross_margin %	units
0	Everlasting Gobstopper	130.00	104.00	80.00	13
1	Fizzy Lifting Drinks	78.75	47.25	60.00	21
2	Fun Dip	12.00	4.80	40.00	8
3	Hair Toffee	76.50	59.50	77.78	17
4	Kazookles	1205.75	92.75	7.69	371
5	Laffy Taffy	53.73	33.48	62.31	27
6	Lickable Wallpaper	7860.00	3930.00	50.00	393
7	Nerds	15.00	7.00	46.67	10

•

Healthy Products

	product_name	business_recommendation
0	Wonka Bar - Milk Chocolate	Healthy Product
1	Wonka Bar - Triple Dazzle Caramel	Healthy Product
2	Wonka Bar - Nutty Crunch Surprise	Healthy Product
3	Wonka Bar -Scrumdiddlyumptious	Healthy Product
4	Wonka Bar - Triple Dazzle Caramel	Healthy Product

9. Division Performance Analysis

Division-level performance evaluation compared:

- Total Revenue
- Total Profit
- Average Margin

```
19
20  -- Division Performance
21  SELECT division,
22         SUM(sales) AS total_sales,
23         SUM(gross_profit) AS total_profit,
24         AVG(gross_margin_percent) AS avg_margin
25  FROM sales_data
26  GROUP BY division;
```

Data Output Messages Notifications

	division text	total_sales double precision	total_profit double precision	avg_margin double precision
1	Sugar	427.48	284.73	57.69
2	Other	9663.25	4333.450000000001	37.671741935483844
3	Chocola...	131692.899999999982	88824.620000000263	67.45710178788748

10. Pareto Analysis

Pareto Analysis was conducted using the 80/20 principle.

Objectives:

- Identify products contributing 80% of revenue

```
#Identify Products Driving 80% Revenue
top_80_revenue_products = pareto_df[
    pareto_df['cumulative_revenue %'] <= 80
]
top_80_revenue_products
✓ [112] 23ms
```

	product_name	sales	gross_profit	revenue_contribution %	cumulative_revenue %
12	Wonka Bar - Triple Dazzle Caramel	28485.00	18610.20	20.090472	20.090472
13	Wonka Bar -Scrumdiddlyumptious	27874.80	19357.50	19.660098	39.750569
10	Wonka Bar - Milk Chocolate	26867.75	17443.37	18.949825	58.700394
9	Wonka Bar - Fudge Mallows	24890.40	16593.60	17.555200	76.255595

- Identify products contributing 80% of profit

```
#Products Driving 80% Profit
top_80_profit_products = profit_pareto[
    profit_pareto['cumulative_profit %'] <= 80
]
top_80_profit_products
✓ [116] 24ms
```

	product_name	sales	gross_profit	revenue_contribution %	cumulative_revenue %	profit_contribution %	cumulative_profit %
13	Wonka Bar -Scrumdiddlyumptious	27874.80	19357.50	19.660098	39.750569	20.715882	20.715882
12	Wonka Bar - Triple Dazzle Caramel	28485.00	18610.20	20.090472	20.090472	19.916141	40.632023
10	Wonka Bar - Milk Chocolate	26867.75	17443.37	18.949825	58.700394	18.667431	59.299454
11	Wonka Bar - Nutty Crunch Surprise	23574.95	16819.95	16.627413	92.883008	18.000263	77.299717

- Detect organizational dependency risks

#Detect Over-Dependency Risk

```
#Count Top Products
num_top_products = len(top_80_profit_products)

total_products = df['product_name'].nunique()

dependency_ratio = (
    num_top_products / total_products
) * 100
```

```
print(dependency_ratio)
```

```
✓ [117] 16ms
```

```
26.666666666666668
```

Findings

A relatively small percentage of products generated the majority of company profit.

This concentration exposes potential business risk if demand declines for these key products.

11. Cost Structure Diagnostics

Cost diagnostics analyzed relationships between:

- Sales
- Cost
- Gross Profit
- Margin Performance

Scatter plots were used to identify:

- Cost-heavy products

- Margin-poor products
- Pricing inefficiencies

```
#Cost vs Sales Scatter Analysis
import matplotlib.pyplot as plt

plt.figure(figsize=(10,6))

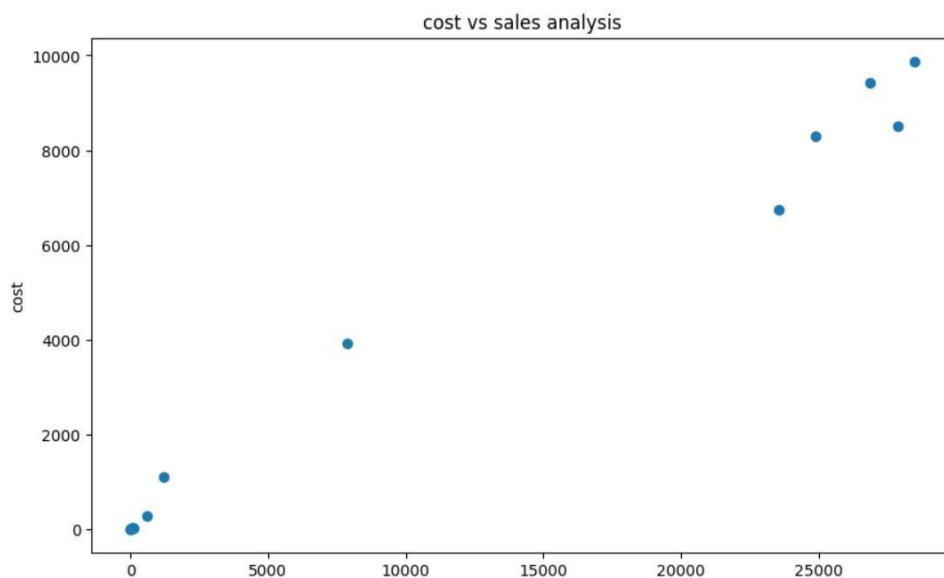
plt.scatter(
    cost_analysis['sales'],
    cost_analysis['cost']
)

plt.xlabel('sales')
plt.ylabel('cost')

plt.title('cost vs sales analysis')

plt.show()
```

✓ [126] 151ms





Business Risk Categories Condition Recommendation

High Cost + Low Margin Cost Renegotiation

High Sales + Low Margin Repricing Required

Low Sales + Low Profit Discontinuation Review

12. SQL-Based Business Analysis

PostgreSQL queries were used to answer critical business questions. **Examples**

- Highest margin products

	product_name	sales	gross_profit	gross_margin %	units
0	Everlasting Gobstopper	130.00	104.00	80.00	13
3	Hair Toffee	76.50	59.50	77.78	17
11	Wonka Bar - Nutty Crunch Surprise	23574.95	16819.95	71.35	6755
13	Wonka Bar - Scrumdiddlyumptious	27874.80	19357.50	69.44	7743
9	Wonka Bar - Fudge Mallows	24890.40	16593.60	66.67	6914
12	Wonka Bar - Triple Dazzle Caramel	28485.00	18610.20	65.33	7596
10	Wonka Bar - Milk Chocolate	26867.75	17443.37	64.92	8267

Division profitability ranking

-- 4. How Profitability Varies Across Product Divisions?

```
SELECT
    division,

    ROUND(SUM(sales)::numeric,2) AS total_sales,

    ROUND(SUM(gross_profit)::numeric,2) AS total_profit,

    ROUND(
        AVG("gross_margin_percent")::numeric,
        2
    ) AS avg_margin,

    SUM(units) AS total_units

FROM sales_data

GROUP BY division

ORDER BY total_profit DESC;
```

	division text	total_sales numeric	total_profit numeric	avg_margin numeric	total_units numeric
1	Chocola...	131692.90	88824.62	67.46	37275
2	Other	9663.25	4333.45	37.67	1242
3	Sugar	427.48	284.73	57.69	137

- Margin-risk products

– 5. Which Products Represent Margin Risk?

```

SELECT
    product_name,

    ROUND(SUM(sales)::numeric,2) AS total_sales,

    ROUND(SUM(cost)::numeric,2) AS total_cost,

    ROUND(SUM(gross_profit)::numeric,2) AS total_profit,

    ROUND(
        AVG("gross_margin_percent")::numeric,
        2
    ) AS avg_margin

FROM sales_data

GROUP BY product_name

HAVING
    AVG("gross_margin_percent") < 40

ORDER BY avg_margin ASC;

```

	product_name text	total_sales numeric	total_cost numeric	total_profit numeric	avg_margin numeric
1	Kazookles	1205.75	1113.00	92.75	7.69

•

Query

Query History

```

2
3  -- Top profitable products
4  SELECT product_name,
5         SUM(gross_profit) AS total_profit
6  FROM sales_data
7  GROUP BY product_name
8  ORDER BY total_profit DESC;
9
10

```

Data Output

Messages

Notifications

≡

+

📄

▼

📋

▼

🗑️

🗄️

⬇️

📈

SQL

Show

	product_name text	total_profit double precision
3	Wonka Bar - Milk Chocolate	17443.36999999991
4	Wonka Bar - Nutty Crunch Surpri...	16819.9499999999757
5	Wonka Bar - Fudge Mallows	16593.600000000013
6	Lickable Wallpaper	3930
7	Wonka Gum	310.70000000000016
8	Everlasting Gobstopper	104
9	Kazookles	92.75
10	Hair Toffee	59.5
11	Fizzy Lifting Drinks	47.25
12	Laffy Taffy	33.48
13	SweetARTS	28.7
14	Nerds	7
15	Fun Dip	4.8

SQL enabled efficient large-scale business analytics and validation of Python-generated insights.

13. Dashboard Development

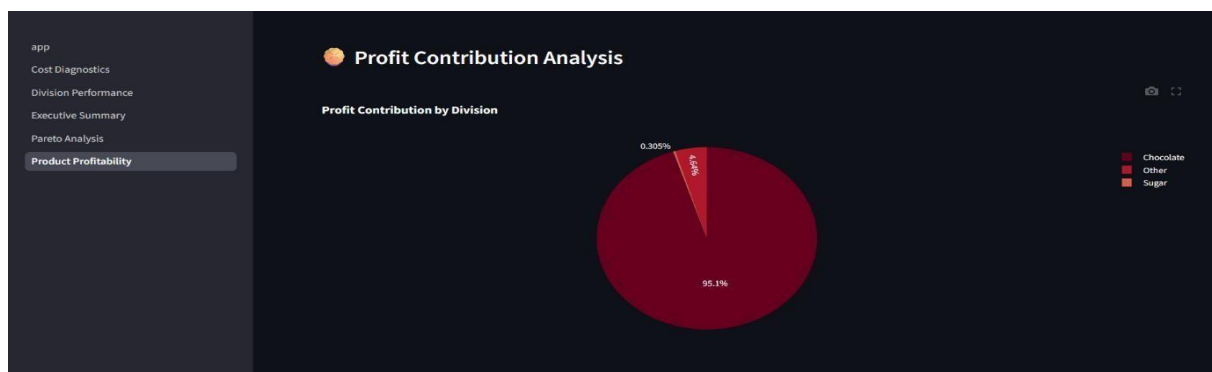
A professional multi-page Streamlit dashboard was developed. **Dashboard Modules**

1. Product Profitability Overview



The dashboard features a sidebar with navigation links: app, Cost Diagnostics, Division Performance, Executive Summary, Pareto Analysis, and Product Profitability (selected). The main content area is titled "Product Margin Leaderboard" and displays a table with 4 columns: product_name, gross_margin_percent, gross_profit, and sales. The table lists 15 products, sorted by gross margin percent.

	product_name	gross_margin_percent	gross_profit	sales
11	Wonka Bar - Nutty Crunch Surprise	71.35	16819.95	23574.95
13	Wonka Bar - Scrumdiddlyumptious	69.44	19357.5	27874.8
9	Wonka Bar - Fudge Mallows	66.67	16593.6	24890.4
12	Wonka Bar - Triple Dazzle Caramel	65.33	18610.2	28485
10	Wonka Bar - Milk Chocolate	64.92	17443.37	26867.75
5	Laffy Taffy	62.31	33.48	53.73
1	Fizzy Lifting Drinks	60	47.25	78.75
14	Wonka Gum	52	310.7	597.5
6	Lickable Wallpaper	50	3930	7860
7	Nerds	46.67	7	15
8	SweetARTS	46.67	28.7	61.5
2	Fun Dip	40	4.8	12
4	Kazookles	7.69	92.75	1205.75



2.

Division Performance Dashboard

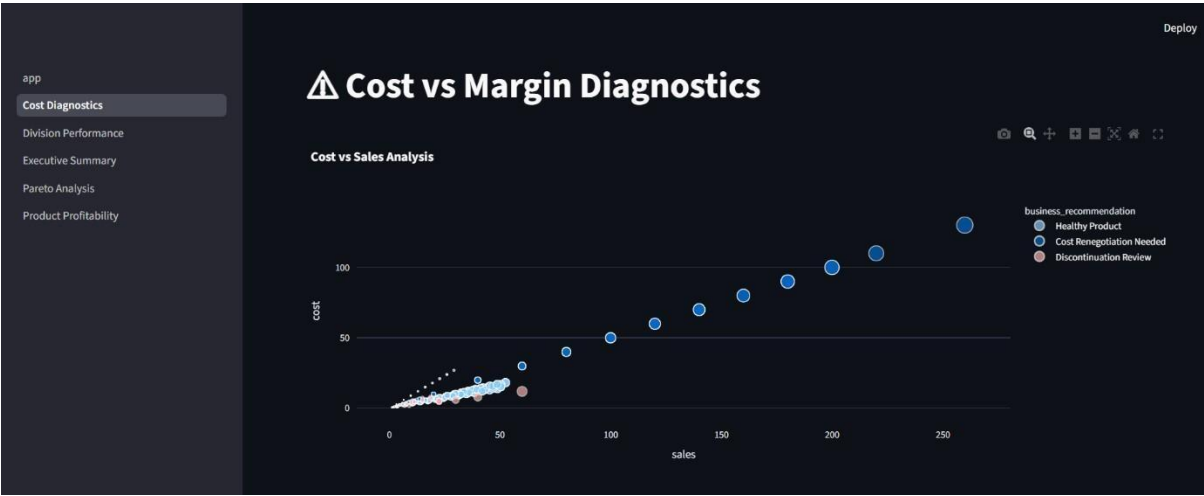


Division Summary

	division	sales	gross_profit	gross_margin_percent	units
0	Chocolate	131692.9	88824.62	67.4571	37275
1	Other	9663.25	4333.45	37.6717	1242
2	Sugar	427.48	284.73	57.69	137

3.

Cost Diagnostics Dashboard



app

Cost Diagnostics

Division Performance

Executive Summary

Pareto Analysis

Product Profitability

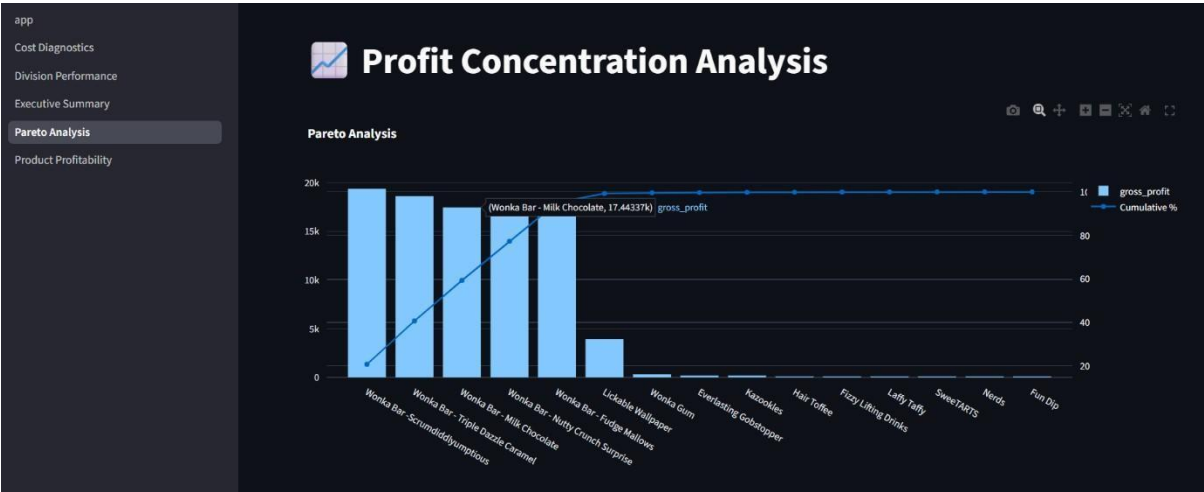
Deploy

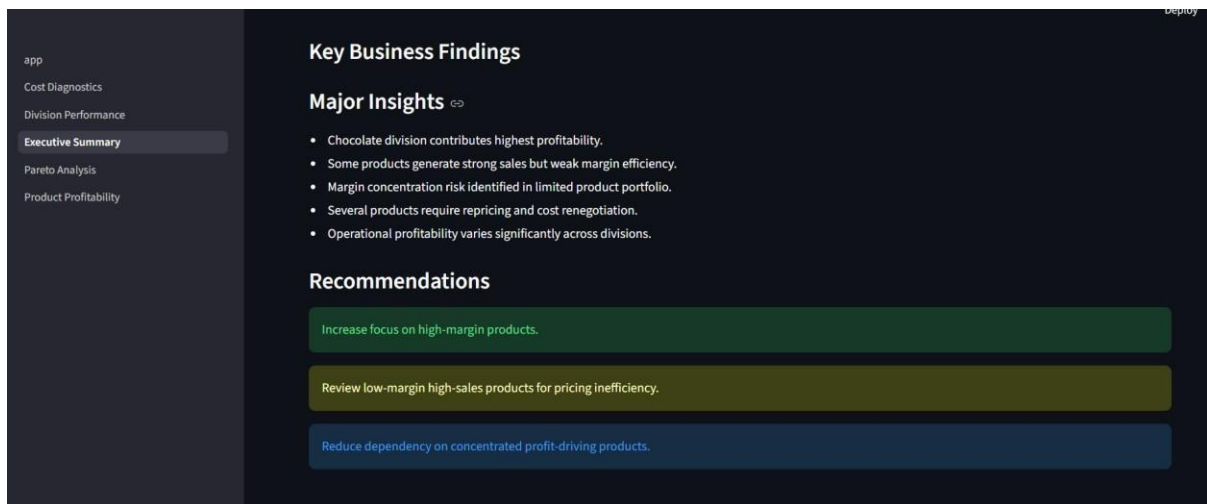
🚨 Margin Risk Products

	product_name	sales	gross_margin_percent	business_recommendation
19	Wonka Gum	1.25	52	Cost Renegotiation Needed
26	Kazookles	9.75	7.69	Cost Renegotiation Needed
32	Wonka Gum	5	52	Cost Renegotiation Needed
92	Kazookles	13	7.69	Cost Renegotiation Needed
99	Wonka Gum	2.5	52	Cost Renegotiation Needed
126	Wonka Gum	3.75	52	Cost Renegotiation Needed
136	Lickable Wallpaper	60	50	Cost Renegotiation Needed
160	Kazookles	19.5	7.69	Cost Renegotiation Needed
187	Wonka Gum	2.5	52	Cost Renegotiation Needed
191	Fizzy Lifting Drinks	7.5	60	Discontinuation Review
198	Wonka Gum	5	52	Cost Renegotiation Needed
200	Kazookles	9.75	7.69	Cost Renegotiation Needed
210	Kazookles	13	7.69	Cost Renegotiation Needed
213	Wonka Gum	8.75	52	Cost Renegotiation Needed
214	Lickable Wallpaper	80	50	Cost Renegotiation Needed
226	Laffy Taffy	11.94	62.31	Discontinuation Review

4.

Pareto Analysis Dashboard





User Features

- Date Range Filter
- Division Filter
- Product Search
- Margin Threshold Slider

The dashboard provides interactive business intelligence for management decisionmaking.

14. Key Insights

The analysis generated several strategic insights:

1. High-sales products are not always highly profitable.
 2. A small group of products contributes the majority of company profit.
 3. Margin performance varies significantly across divisions.
 4. Several products exhibit pricing inefficiencies.
 5. Cost-heavy products reduce profitability despite strong sales.
-

15. Business Recommendations

Based on analytical findings, the following recommendations were proposed:

Repricing Strategy

Increase pricing for high-sales low-margin products.

Cost Optimization

Negotiate supplier costs for margin-poor products.

Product Rationalization

Review low-performing products for discontinuation.

Focus on High-Margin Products

Allocate resources toward products demonstrating superior profitability. **Risk**

Diversification

Reduce over-dependency on a small group of highly profitable products.

16. Future Enhancements

Future versions of the project may include:

- Sales Forecasting
 - AI-Generated Insights
 - Factory Location Mapping
 - Cloud Deployment
 - Power BI Integration
 - Customer Segmentation
 - Predictive Analytics
-

17. Conclusion

This project successfully transformed transactional sales data into actionable profitability intelligence for Nassau Candy Distributor.

By integrating Python, PostgreSQL, and Streamlit, the analysis provided a comprehensive understanding of product profitability, division performance, cost efficiency, and margin risk.

The resulting insights support data-driven pricing decisions, cost optimization initiatives, and strategic portfolio management. The dashboard further enables stakeholders to monitor performance interactively and make informed business decisions based on real-time analytical insights.

References

- [Python Documentation](#)
- [PostgreSQL Documentation](#)
- [Streamlit Documentation](#)
- [Plotly Documentation](#)
- [Pandas Documentation](#)
- [Business Intelligence and Profitability Analysis Literature](#)